

Answer **all** questions.

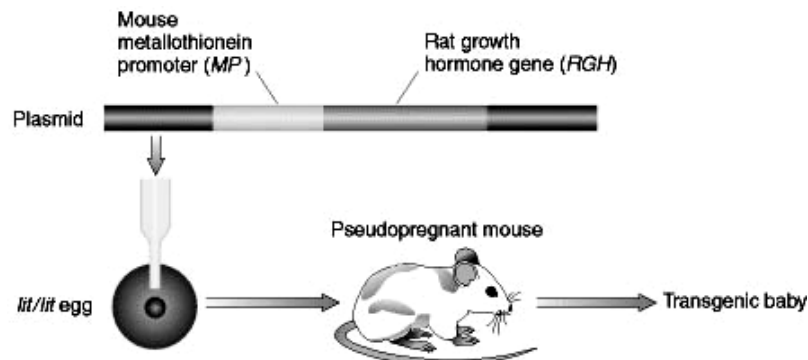
1. The first example of gene therapy in a mammal was the correction of growth hormone deficiency in mice. The recessive mutation *little* (*lit*) results in dwarf mice. Even though a mouse's growth hormone gene is present, growth hormone is not produced.

(a) Describe how the *little* mutation resulted in dwarfism.

[3]

1. *little* mutation is a nonsense mutation;
2. substitution of a codon to a stop codon;
3. both alleles of growth hormone gene are mutated;;
4. subsequently, growth hormone polypeptide would not be translated;
5. there would not be functional growth hormone to stimulate growth;

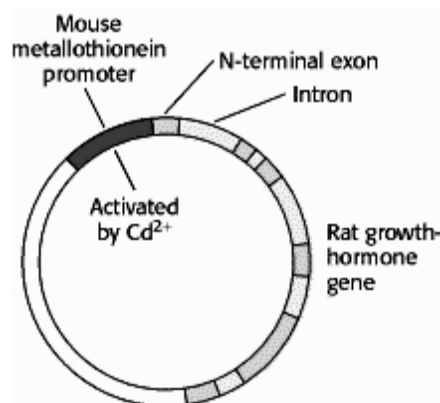
The initial step in correcting this deficiency was to inject homozygous *lit / lit* eggs with a 5-kb plasmid that contained the rat growth hormone structural gene (*RGH*) fused to a promoter sequence from the mouse metallothionein gene (*MP*). The normal role of the metallothionein protein is to detoxify heavy metals, hence the promoter sequence is responsive to the presence of heavy metals in the animal. The eggs were then implanted into pseudopregnant mice, and the transgenic baby mice were raised as shown in **Fig. 1.1**.



Source: *Modern Genetic Analysis*. Griffiths AJF, Gelbart WM, Miller JH, et al. New York: W. H. Freeman; 1999.

Fig. 1.1

Fig. 1.2 shows the 5-kb plasmid that contained the *MP-RGH* gene construct. The *RGH* gene was inserted next to the *MP*, which is activated by the addition of heavy metals, such as cadmium ions to the drinking water.



Source: *Biochemistry*. 5th edition. Berg JM, Tymoczko JL, Stryer L. New York: W H Freeman; 2002.

Fig. 1.2

1. (b) Suggest the advantage of fusing the mouse metallothionein promoter (*MP*) to the rat growth hormone structural gene (*RGH*). [2]

1. addition of heavy metals; would induce the transcription of *RGH* gene;
2. this allows for control / regulation / activation of *RGH* gene expression; at the appropriate developmental stage;

- (c) Suggest why it is not necessary to include selectable markers in the plasmid. [2]

1. homozygous *lit/lit* mice used for microinjection are dwarfs;
2. upon successful transfection, *RGH* gene would be expressed;
Accept: functional *RGH* protein would be expressed in the transgenic baby mice;
3. this confers a well-defined trait / observable phenotype on the transgenic baby mice;
4. i.e. increased size / weight;

Current treatment of human dwarfism involves the injection of human growth hormone (hGH), which is expressed by genetically engineered bacteria, into the bloodstream of patients.

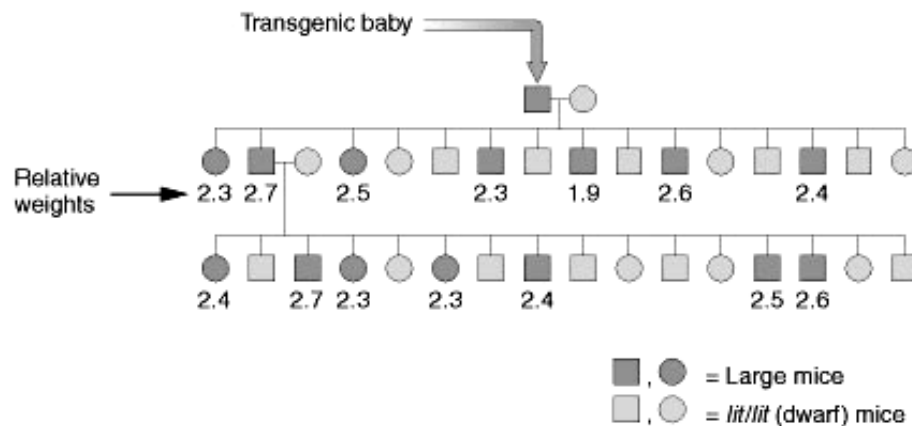
- (d) With reference to **Fig. 1.2**, explain the difference between the plasmid used in gene therapy and the plasmid used in bacterial transformation. [2]

1. *GH* gene used for gene therapy contained introns; but that used for bacterial transformation does not;
2. mouse is a eukaryotic organism;
3. which contains the splicing mechanisms;
4. that are necessary for splicing out the introns and splicing together the exons;
Accept: reverse arguments wrt bacteria

- (e) Describe how the human growth hormone (*hGH*) gene is processed to make it compatible for expression in bacteria. [3]

1. extract mRNA from pituitary gland;
2. introns have been spliced out of mature mRNA;
Accept: mature mRNA not interrupted by introns
3. exons that have been spliced together are continuous; such that
4. codons can be read in the proper frame by the bacterial ribosome;
5. use reverse transcriptase; to catalyse formation of double-stranded DNA;
6. using mRNA as the template;

1. A representative transgenic mouse was crossed with a homozygous *lit / lit* female. The resulting pedigree is shown in **Fig 1.3**: mice two to three times the weight of their *lit / lit* relatives are produced in subsequent generations.



Source: *Modern Genetic Analysis*. Griffiths AJF, Gelbart WM, Miller JH, et al. New York: W. H. Freeman; 1999.

Fig. 1.3

- (f) With reference to **Fig. 1.3**, explain why the *RGH* transgene is inherited in a dominant fashion. [2]
 1. representative transgenic mouse is heterozygous;
 2. representative transgenic mouse carries only one copy of the *RGH* transgene;
 3. heterozygous mice express the phenotype;
 4. i.e. demonstrated at least two-fold weight increase; compared with *lit / lit* relatives
- (g) Discuss a concern of applying the abovementioned gene therapy to human subjects. [1]
 1. use of heavy metals; e.g. cadmium ions being added to drinking water to induce expression of *RGH* gene
 2. reference to heavy metal poisoning;

[Total: 15]

2.

PART I

An egg cell from a mouse was fertilised in a laboratory and allowed to develop into an eight-celled embryo. This was split into four parts, each of which developed into a new embryo, as shown in **Fig. 2.1**. The new embryos were later transferred into different surrogate mice.

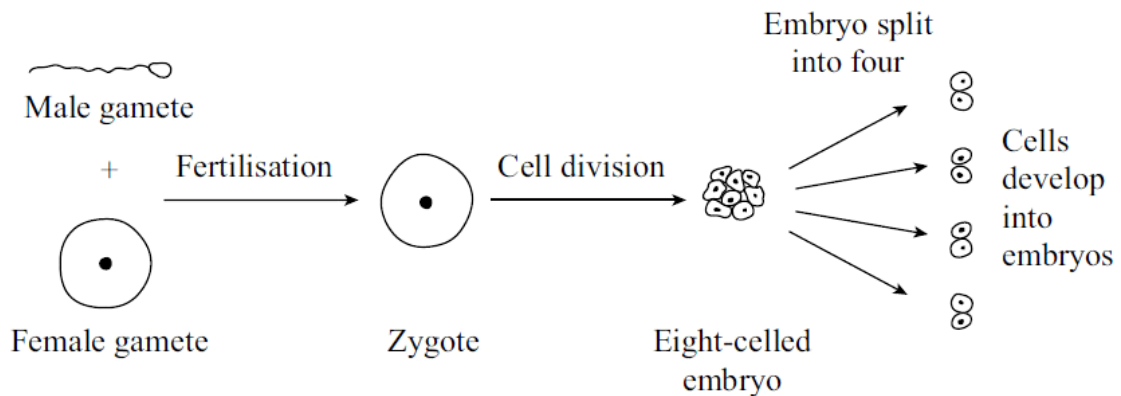


Fig. 2.1

(a) Explain why the new embryos produced are clones. [2]

1. cells are produced by mitosis;;
2. cells are genetically identical;;

(b) If embryos with more than eight cells are split up, the separated cells fail to develop into new embryos. Suggest why the separated cells fail to develop into new embryos. [2]

1. cells are no longer totipotent;;
Accept: cells have differentiated / become specialised
2. through selective changes; in genetic activity;
Accept: some genes are turned off and others are turned on

PART II

In an animal model of cystic fibrosis, normal *CFTR* alleles were transferred into tracheal epithelial cells of affected adult mice using liposomes shown in **Fig. 2.2**.

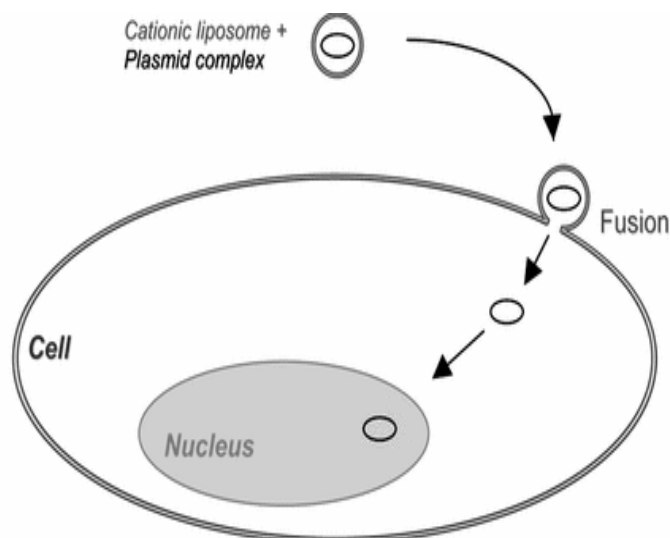


Fig. 2.2

2. (c) Explain why affected adult mice that have been treated by this method do **not** pass on the normal *CFTR* allele to their offspring. [1]
1. reference to tracheal epithelial cells being somatic cells;
 2. gametes do not contain *CFTR* allele;
- (d) With reference to **Fig. 2.2**, discuss the limitations of using liposomes as a gene delivery system for the normal *CFTR* allele. [2]
1. *CFTR* allele delivered does not integrate into chromosomes; → each time tracheal cell divide, delivered *CFTR* allele are lost;
 2. Only a fraction of tracheal epithelial cells; fuse with *CFTR*-containing liposomes;
- (e) Viruses have been used to transfer the normal *CFTR* allele into murine tracheal epithelial cells. Suggest **two** advantages for using viruses to transfer the therapeutic allele into cells. [2]
1. Therapeutic allele can be integrated into the chromosomes; i.e. can be replicated together with the chromosome prior to mitosis; allowing for long-term, stable expression of *CFTR* protein;
 2. Therapeutic allele can be targeted; to the tracheal epithelial cell;

[Total: 9]

3. Genetic engineering is a technique that has been developed to improve the quality and yield of many crops plants.

(a) Explain the advantages of tissue culture in cloning genetically modified plants. [2]

1. Rapid multiplication of plants, giving rise to large no. of GM plants;;
2. Produce genetically identical individuals and possess the desirable features of the stock (GM) plant;;

A number of different crop plants have been genetically engineered to express a gene for an insecticidal toxin (*Bt* toxin) from a bacterium, *Bacillus thuringiensis*, that kills many insect species.

In China, *Bt* cotton has been grown since 1997. A survey at the end of 2001 showed that it was being grown by over two million farmers on fields totalling more than 7000 km².

Further findings of the survey are shown in **Table 3.1**.

Table 3.1

survey finding	percentage of reported cases of insecticide poisoning among cotton farmers	cost of producing 1kg of cotton / US \$
farmers growing non- <i>Bt</i> cotton	22	2.23
farmers growing <i>Bt</i> cotton	5	1.61

(b) Describe and explain the findings of the survey. [3]

ref to figures can only be awarded if data is manipulated – do not award for simply quoting the data from the table

1. *Bt* cotton produces *Bt* toxin which kills larvae of insect;
 2. no / less, insecticide needed;
 3. reduced number of cases of pesticide poisoning;
 4. ref to figures (e.g. by x 4.4);
 5. reduced cost (insecticide);
 6. ref to figures (e.g. by 0.62 US\$ kg⁻¹ / x 1.38);
- A: reverse arguments*

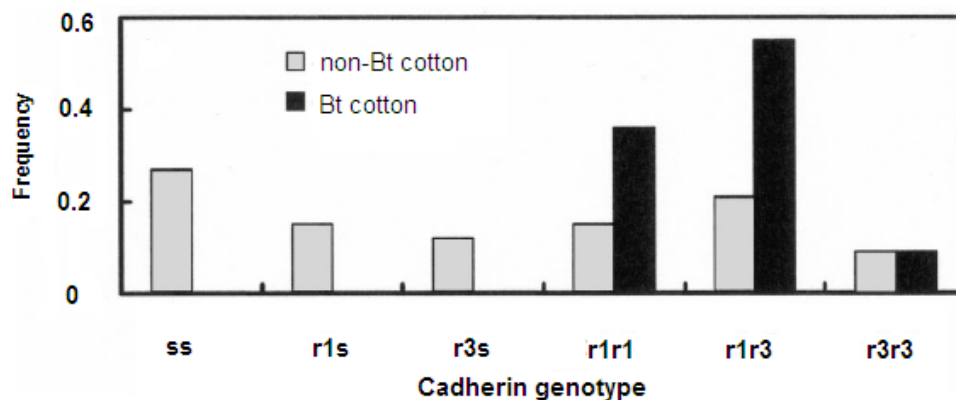
The *Bt* toxin binds to cell membranes of the cells in the insect gut. The receptor for the toxin has been identified as a membrane-anchored protein encoded by the cadherin gene. *Bt* toxin forms ion channels that cause an efflux of potassium ions from the insect gut cells, leading to cell lysis.

The cotton bollworm, *Helicoverpa armigera*, is one of the most serious insect pests of cotton in Asia, Australia, and Africa. A major deletion of the cadherin gene was associated with a high level of *Bt* toxin resistance in *H. armigera*. Three alleles (*r1*, *r2*, and *r3*) of the cadherin gene in the resistant strain have been identified:

- *r1* allele lacks 24-bp
- *r2* allele has a 202-bp deletion
- *r3* allele has a 126-bp deletion

3. (c) (i) Explain why mutations of the cadherin gene could confer resistance to *Bt* toxin in *H. armigera*. [2]
1. deletion leads to removal of corresponding mRNA codons;
 2. removal of amino acids in the polypeptide chain;
 3. change in protein conformation/ affect folding of polypeptide chain;
 4. unable to bind to *Bt* toxin;
 5. prevent cell lysis
- (ii) Explain why gel electrophoresis can be used to distinguish the three alleles. [2]
1. Difference in size for the three alleles ;;
 2. DNA fragments are separated on the basis of size;
 3. i.e. shorter DNA fragments migrate through the pores of the agarose gel faster (i.e. more readily) than longer DNA fragments;; or vice versa
- (d) A survival test was conducted on larvae of different genotypes to assess their resistance to *Bt* toxin.

* s allele refers to the normal / susceptible allele



Source: Association Between Resistance to *Bt* Cotton and Cadherin Genotype in Pink Bollworm. Tabashnik et al. *J. Econ. Entomol.* 98(3): 635-644 (2005)

Fig. 3.1

State and explain the nature of the mutant alleles that confer resistance to *Bt* toxin. [3]

1. Recessive;;
 2. presence of only 1 mutant allele, *r1* or *r3* , is not able to confer resistance to *Bt* toxin;;
 3. evidence : *r1s* or *r3s* → none survive on *Bt* cotton;;
- OR
2. required the presence of 2 mutant alleles to confer resistance to *Bt* toxin;;
 3. evidence : *r1r1*, *r1r3*, *r3r3* → some larvae survive on *Bt* cotton;;

3. (e) Concerns about resistance to *Bt* crops led the U.S. Environmental Protection Agency to mandate the “refuge strategy” for delaying pest resistance. Thus, farmers growing cotton that produces *Bt* toxin must plant at least 20% of non-*Bt* cotton in the field.

Explain how the “refuge strategy” could delay pest resistance to *Bt* toxin in an insect population. [2]

1. enable survival of susceptible pests;
2. resistant pest can mate with susceptible pests;
3. retain susceptible allele in the population/ offspring will still be susceptible to *Bt* toxin/ reduce the selection pressure for resistance pests;

- (f) Outline two possible hazards of genetically engineered crop such as *Bt* cotton. [2]

1. pass resistance to other weeds ;;
2. kill non-target insects which are beneficial ;;
3. ref to potential effect on food chain ;;
4. ref to potential effect on biodiversity ;;

[Total: 16]

Planning Question

4. Whole algal cells can be added to the sodium alginate and immobilised in the gel beads as shown by the procedure in **Fig. 4.1**.

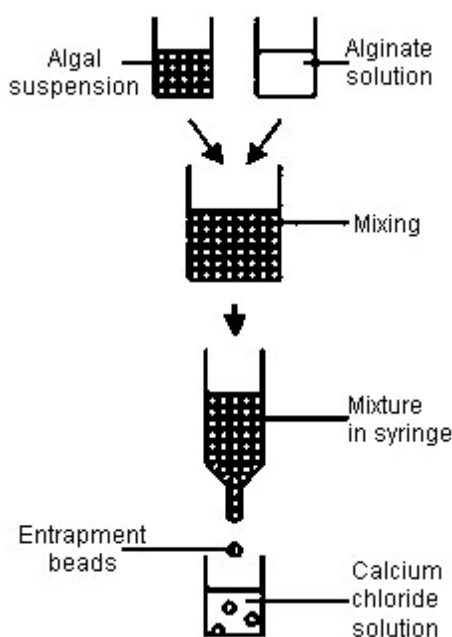


Fig. 4.1

Scenedesmus quadricauda is a unicellular, photosynthetic green alga that absorbs carbon dioxide for photosynthesis.

Hydrogen carbonate indicator is very sensitive to changes in carbon dioxide levels. The indicator is orange / red when equilibrated with atmospheric carbon dioxide. The colour changes (i) to yellow when more carbon dioxide is added; and (ii) through red to purple when carbon dioxide is removed.

Design an experiment, using algal cells immobilised in alginate beads, to test the hypothesis that:

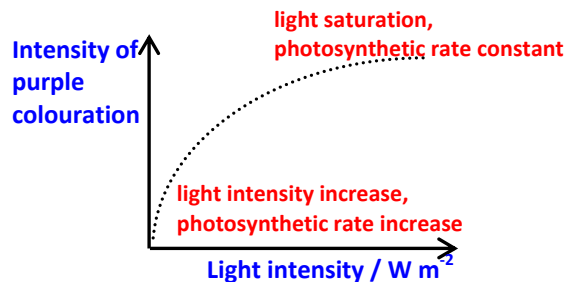
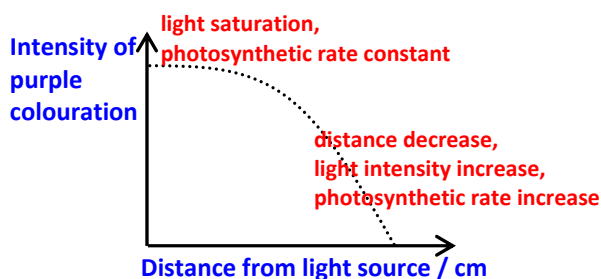
The rate of photosynthesis is dependent on light intensity. **[12]**

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method,
- an explanation of the dependent and independent variables involved,
- relevant, clearly labelled diagrams,
- how you will record your results and ensure they are as accurate and reliable as possible,
- proposed layout of results tables and graphs with clear headings and labels,
- the correct use of technical and scientific terms,
- relevant risks and precautions taken.

Introduction i.e. explanation of the dependent and independent variables involved [max 3]:

1. photosystems in thylakoid membrane of chloroplasts are involved in the harvest of light energy;
2. harvested light energy:
 - a. is used to displace an electron from reaction centre chlorophyll *a* in PSI or PSII;
 - b. is also involved in the photolysis of water to replace electron displaced from PSII;
3. ETC in thylakoid membrane harness energy from transition of electrons to pump protons from stroma into thylakoid space;
4. ATP synthase complex in thylakoid membrane couples energy from diffusion of protons down concentration gradient to the phosphorylation of ADP;
5. electrons from ETC are used to reduce NADP⁺;
6. light intensity can be achieved by varying distance between light source and experiment; i.e. as distance ↑, light intensity ↓;
7. relate ↑ light intensity to ↑ light reaction to ↑ production of ATP and NADPH/H⁺; provided other potential limiting factors are controlled
8. explain that (i) ATP is used to activate 3-phosphoglycerate to 1,3-bisphosphoglycerate and (ii) NADPH/H⁺ is used to reduce 1,3-bisphosphoglycerate to 3-phosphoglyceraldehyde (PGAL);
9. relate to ↑ CO₂ fixation to ↑ CO₂ absorption to ↑ intensity of purple colouration of hydrogen carbonate indicator;
10. Sketch of graph: trend at low light intensity; light saturation; [max 1]



Under methodology: [max 2]

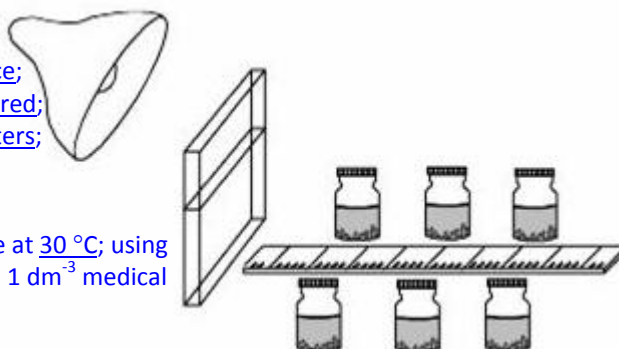
11. independent variable : light intensity; vary distance between light source and algal beads e.g. 10 cm, 20 cm, 30 cm, 40 cm, 50 cm, 60 cm, and 70 cm
12. dependent variable : intensity of purple colouration; which is determined by colourimeter
13. controlled variable :
 - a. power rating of light source
 - standardise at 150 W;
 - b. wavelength of light
 - standardise at blue / red; using photographic filters;
 - c. temperature
 - standardise at 30 °C; using heat screen, e.g. 1 dm⁻³ medical flats;
 - d. amount of algal cells
 - standardise using same number; size / diameter; of algal beads
 - e. amount of CO₂
 - standardise using same volume;
 - glass containers with lids must be of same size;

Method i.e. description of the method used including the scientific reasoning behind the method / relevant, clearly labelled diagrams: [max 3]

14. rinse with a small volume of hydrogen carbonate indicator; 21 small glass containers with lids;
15. add 20 algal beads, each 3 mm in diameter; to each glass container
16. add 20 cm³ of hydrogen carbonate indicator to each glass container; i.e. to ensure that the indicator solution covers the algal beads; and replace the lid;
17. place the containers at different distances from the light source; i.e. 10 cm, 20 cm, 30 cm, 40 cm, 50 cm, 60 cm, and 70 cm;
18. ensure that there are 3 replicates for each distance from the light source;

- 13 a. use 150 W light source;
- 13 b. standardise at blue / red;
- using photographic filters;

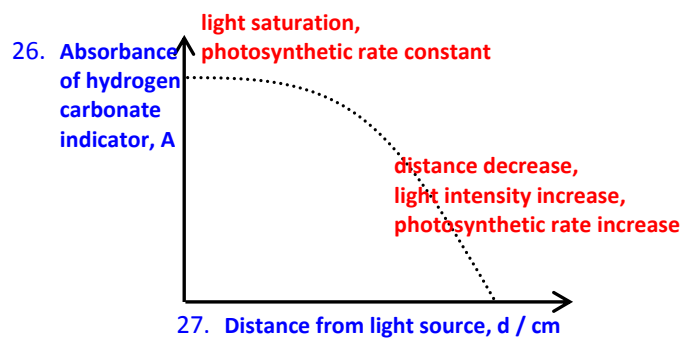
- 13 c. standardise at 30 °C; using heat screen, e.g. 1 dm⁻³ medical flats



19. leave the set-up for 1 hour;
20. at the end of 1-h incubation, fill a cuvette with 3 cm³ of deionised water and place in the colourimeter. Press the zero button;
21. fill a cuvette with 3 cm³ of hydrogen carbonate indicator from one of the test solutions. Place the cuvette in the colourimeter. Press the test button and take the reading; Repeat for each of the test solutions.
22. Repeat the entire experiment three times;
23. Conduct a negative control;
 - a. which is subjected to the same factors; (i.e. 20 algal beads each 3 mm in diameter, 20 cm³ of hydrogen carbonate indicator, covered glass bottle placed 10 cm away from light source) as that for the experiment, except that light source is switched off;
 - b. it would be observed that hydrogen carbonate indicator remains orange/red; this proves that it is indeed light (and not any other factors) that causes the observed phenomenon;

Expected Results i.e. how you will record your results and ensure they are as accurate and reliable as possible, proposed layout of results tables and graphs with clear headings and labels: [max 2]

24. Distance from light source, d / cm	25. Absorbance of hydrogen carbonate indicator, A			
	A ₁	A ₂	A ₃	A _{average}
10				
20				
30				
40				
50				
60				
70				



Safety Precaution i.e. relevant risks and precautions taken:

28. Hydrogen carbonate indicator solution is flammable. Avoid open flame.
29. Calcium chloride solution can be an irritant. Wear goggles.
-

Free-response question

Write your answer to this question on the separate answer paper provided.

Your answer:

- should be illustrated by large, clearly labelled diagrams, where appropriate;
- must be in continuous prose, where appropriate;
- must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

A **NIL** return is necessary if you have not attempted this section.

5. With reference to the analysis of DNA molecules, describe and explain the use of

(a) polymerase chain reaction (PCR); [8]

- A1 3 step process – denaturation then annealing then extension/ elongation;
A2 ref 94-96°C (denaturation) then 45-60°C (annealing) then 72°C (elongation);
A3 carried out in thermal cycler;

Denaturation

- A4 hydrogen bonds holding strands together break;
A5 forming single stranded DNA;

Annealing

- A6 complementary primer attaches to single stranded DNA/DNA template;
A7 ref. forward and reverse primers;

Elongation/Extension

- A8 *Taq* DNA polymerase attaches to the primer and starts copying template/ adding nucleotides/ polymerising DNA;
A9 complementary nucleotides join to form strand/ extend strand attached to primer
A10 catalyses phosphoester bonds between adjacent nucleotides;
A11 add nucleotides to 3' ends on both primers /extend in the 5' → 3' direction;
A12 process repeats itself to produce multiple copies of target sequence;
max 5

Use of PCR

- A13 amplifying minute amount of DNA;
A14 producing specific gene (fragment);
A15 much faster than previous method using bacteria;
A16 ref cloning genes for use in genetic manipulation;
A17 ref DNA fingerprinting / detect trace amount of DNA from tissues at crime scene /
A18 ref diagnose genetic conditions (using DNA from embryonic cells);
A19 paternity testing;
A20 ref studying human ancestry / studying ancient DNA from fossil remains;

max 3

(b) gel electrophoresis;

[6]

- B1 loading of DNA fragments at one end of the agarose gel;
- B2 electric field or voltage applied across the gel;
- B3 DNA being negatively charged;
- B4 will move towards the positive electrode/anode;
- B5 rate of movement/speed/distance travelled in a given time depends on the size of DNA fragments;
- B6 larger DNA fragments travel slower and smaller fragments traveling faster;
- B7 more resistance for the larger fragment to move through the pores of the gel;
- B8 DNA ladder is added to calibrate;
- B9 use of ethidium bromide / fluorescent dyes and UV light to see the DNA bands;

max 4

Use of Gel electrophoresis

- B10 identification of different alleles/ mutation of genes ;;
- B11 separation of protein subunits ;;
- B12 DNA Fingerprinting/Restriction Fragment Length Polymorphisms ;;
- B13 AVP;;

max 2

(c) restriction fragment length polymorphism (RFLP) analysis.

[6]

- C1 use of specific restriction enzymes to cut DNA into fragments/short lengths;
- C2 specificity of restriction enzyme cutting sites;
- C3 different restriction fragments of different lengths are produced;
- C4 (Polymorphisms are) the variety of human genotypes/inherited genetic differences among individuals in a population/the varied alleles of the genes in the gene pool;
- C5 IDEA OF polymorphisms can be caused by any change to base sequence/even a change to a single nucleotide;

Use of RFLP

- C6 screening for the presence of disease-causing alleles;
- C7 ref. use before conception to screen parent's DNA / use after amniocentesis to screen fetal DNA;
- C8 DNA fingerprinting;
- C9 ref. to VNTR (variable number of tandem repeats) or STR (short tandem repeats);
- C10 identify criminals / determine paternity;
- C11 genomic mapping;
- C12 use RFLP markers as genetic markers for a particular location/locus in the genome;
- C13 ref. frequency of recombination between 2 RFLP markers indicate the distance between two loci on a chromosome;

[Total: 20]